

# Jack Brewer

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## EDUCATION

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### University of Bristol

Computer Science MEng; Projected 2:1

Sep. 2023 — Jun. 2027

## PROJECTS

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### Optimisation of a Fluid Simulation, C, OpenMP, MPI

Jan. 2026 — May 2026

- Optimised a D2Q9 Lattice-Boltzmann fluid simulation, achieving a  $3\times$  single-thread speed-up and  $153\times$  across a 144-core Isambard 3 node, then  $407\times$  with MPI across four nodes (576 cores)
- Compared GCC and NVIDIA nvc, with and without vectorisation, at each stage of optimisation. Found the best compiler and flags changed once vectorisation was introduced

### Functional Verification of a Calculator Design, e (Specman)

Sep. 2025 — Dec. 2025

- Built a black-box Specman testbench for a four-port calculator design, with a constrained random driver, reference model, scoreboard-style checker and coverage collector, plus a regression script that merges coverage across runs
- Reached 93.4% code coverage and documented 8 bugs, each with a reproducing test, including port starvation under load, missing overflow flags on ports 2 and 3, and corrupted left-shift results

### Caligula - Games Project, C#, Unity, Arduino, BLE

Jan. 2026 — May 2026

- Managed a team of seven building a two-player co-operative hack-and-slash game with custom hardware, organising sprint meetings, task allocation, and communication with supervisors and technology experts
- Built a custom motion controller (BNO085 IMU, RP2040) that streams orientation and acceleration over BLE to an Arduino hub, which appears to the PC as a standard COM port to work around Windows 11's lack of native BLE support
- Integrated the controller into Unity with a buffered, multithreaded C# serial reader, and fixed latency and data corruption by moving from software to hardware serial and tuning baud rate and scan intervals

### Graphics Engine, C++

Sep. 2025 — Dec. 2025

- Built a 3D graphics engine from scratch in C++ and SDL with raytraced and rasterised modes
- Implemented transformations, texture mapping, lighting and camera movement from the underlying linear algebra and used OpenMP to parallelise raytracing mode

## WORK EXPERIENCE

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### Graduate Teacher Level 1 & 2

Sep. 2025 — Present

University of Bristol

Bristol

- Software Engineering Project - Mentored three groups of students acting as a critical point of contact to discuss project progress and challenges. Consulted in workshops and one-on-one sessions for less confident students to become proficient with technologies such as Git and Docker
- Design Verification - Helped students understand verification and validation techniques and with navigating and using software such as QuestaSim, Specman and IMC
- Computer Graphics - Helped students debug C++ raytracer and rasteriser
- Imperative Programming - Supported new students in setting up and understanding Linux environments, understanding imperative concepts and debugging C

### Student Developer - Airship Game | JavaScript

May 2025 — Sep. 2025

University of Bristol

Bristol

- Audited a large, undocumented JavaScript codebase used by hundreds of prospective students a year
- Identified usability and documentation gaps and rewrote the user documentation to match current behaviour

## OTHER QUALIFICATIONS

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### Outdoor Pursuits 16 Hour First Aid Level 3 (VTQ), ProTrainings

Valid 28 Sep. 2024 - 27 Sep. 2027

## SKILLS

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- Programming Languages** (Proficient): C#, TypeScript, C, C++
- Programming Languages** (Basic): e Language (Specman), JavaScript, Python
- Technologies**: NestJS, Git, SQLite, Docker, OpenMP, MPI, Linux, Unity